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## plant detection final

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# ViraLeaf : Smarter Leaves. Healthier Crops.

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***Abstract*— Plant diseases have a great influence on agricultural productivity as they lower the quality and quantity of crop production. Early and precise diagnosis of foliage diseases is a crucial aspect in promoting healthy growth in plants and enhancement of food security. The proposed project introduces a machine learning-based plant disease detection system, which works on the principles of image processing and machine learning to classify infected leaves and determine the percentage of disease on the leaf.Moreover, the model which is proposed does not only identify the presence of disease but also determines the level of severity of the disease depending on the area of the leaf which is infected. By determining the percentage of infection, farmers will be in a position to implement preventive measures in a timely manner including the application of relevant pesticides, pruning of affected leaves or an alternative method of irrigation to prevent the disease. The method aids in reducing crop wastage, enhancing yield, as well as aiding precision agriculture by means of effective and smart disease tracking.**

***Keywords—Plant Disease Detection, Image Processing, Learning, Leaf Disease Classification, Disease Severity estimat ion, Precision Agriculture.***

## I. Introduction

Agriculture is significant in sustaining the world economy and food security to the ever-increasing population. Nonetheless, diseases of plants are also a big issue that lowers the quality and quantity of crops. Diseases of leaves are extremely widespread and have a direct impact on photosynthesis and development of the plant. When they are not detected at their early stages, the diseases might spread rapidly and result in severe loss of crops hence loss of money by farmers.

The farmers or experts are traditionally known to identify plant diseases by manual observation of the leaves. This is a very time consuming process and experience is necessary to diagnose the disease correctly. In big farmlands, it will be hard to trace all the plants and the probability of inaccuracies is high. Due to these shortcomings, there is the necessity to have an automated system which can help identify plant diseases without involving human services fully and without having to rely entirely on human services, in order to ensure that the system is quick and accurate in identifying the disease.

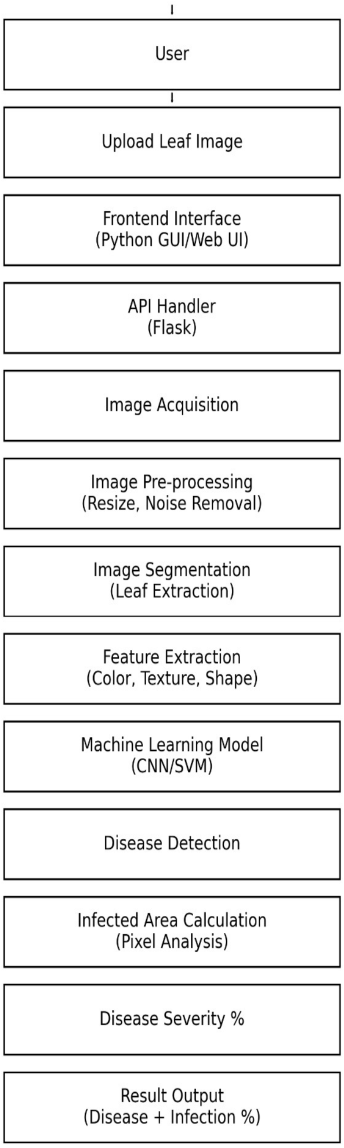
As the techniques of image processing and machine learning have improved, now it is possible to analyze the images of leaves and recognize the diseases using such features as colour, texture and shape. The proposed project will introduce a system that will not only identify diseases in plant leaves, but also analyze the percentage of the affected area in order to establish how serious a disease is. The system also offers appropriate remedies, as it assists the farmers to make appropriate actions to conserve crops, enhance productivity and seek improved and sustainable farming procedures.

## II. Literature Review

Much has been done in terms of detecting plant diseases through image processing and machine learning. Primarily, the initial studies by Revathi and Hemalatha [1] had used image edge detecting techniques to diagnose cotton leaf spot diseases. Arivazhagan et al. [2], [3] suggested the methods of identifying the unhealthy areas of the plant leaves based on the texture features, which allows distinguishing between the plant diseases. Barbedo [4] gave several digital image processing methods in detecting, quantifying and classifying the disease of plants, with emphasis put on feature extraction.

Various articles have been aimed at enhancing the precision and effectiveness of disease diagnosis. The system introduced by Al-Hiary et al. [5] was a quick and precise system based on the image segmentation methods. Gavhale and Gawande [6] presented the general picture of the image processing-based methods and underlined their efficiency in agriculture. Pydipati et al. [8] and Camargo and Smith [7] investigated the color and texture-based classification methods to detect plant diseases.

The recent studies have switched to the more powerful machine learning and deep learning models. Sladojevic et al. [9] proposed the use of deep neural networks in automatic recognition of plant diseases by images of leaves. To enhance the accuracy of detection, Singh and Misra [10] used image segmentation and soft computing methods. Convolutional Neural Networks (CNN) were found by Ferantinos [11] to be highly accurate in the detection and diagnosis of plant diseases.




Even though these studies prove the efficiency of image processing and deep learning methods, the majority of available systems are oriented primarily on the disease classification and do not predict the severity of infection. Also, a large portion of systems do not have practical applications like remedy suggestions and real time user-friendly interfaces to be used in agriculture.

The system proposed overcomes these shortcomings through the combination of image processing and machine learning to not only identify the disease of plant leaves but also the percentage of the affected area. In addition, the system offers organic and inorganic solutions, which makes it an effective and convenient solution to the farmers to monitor the health of the plants and take the necessary preventive measures in time.

## III. Proposed Methodology

The given Plant Leaf Disease Detection System is created to help automatically recognize the presence of plant diseases and the extent of their impact with the help of image processing and machine learning tools. The system adheres to an orderly work process which has several phases, beginning with input of images up to production of final results.

The user will first give an image of a leaf using the interface. This picture is the input of the system. After acquiring the image the image is then sent to the preprocessing stage where the image is resized to a standard size and noise eliminated by filtering techniques. Image enhancement is also carried out to enhance clear images and bring out some important features necessary to analyze them..

Once prepossessed, image segmentation is used to separate the leaf and the background and detect areas of infection that are present on the surface of the leaf. The step is significant in isolating the diseased part of the leaf that may be used to enhance the precision of detection. The segmented image is able to clearly differentiate between healthy and infected regions.

Following this, an extraction of features is done to derive important properties like color variations, texture patterns and a difference in shape based on the image of the segmented leaf. These characteristics are important in the diagnosis of the disease. The features extracted are subsequently provided as input to a machine learning model e.g. Support Vector Machine (SVM) or Convolutional Neural Network (CNN), which identifies the leaf as a healthy or diseased.

When the leaf is found to be diseased, the system goes ahead to estimate the area that is infected by analysing the pixels. The percentage of infected pixels is divided by the sum of pixels in the leaf area to give the percentage infection. This percentage can be used to estimate the level of severity of the disease.

The last step is where the system shows the result with the name of the disease that has been detected, the percentage of infection and the level of severity. The system, also, offers appropriate organic and inorganic solutions to manage the disease. This enables the farmers and the users to undertake timely preventive measures and to enhance crop yield.

## IV. Experimental Results

### A. Experimental Setup

Plant Leaf Disease Detection System is a proposed system that was created and experimented with Python and image processing, machine learning. The data comprises various images of leaves that have normal and diseased leaves. All the input images were pre-processed by resizing them to a standard size and normalizing them. The system was tested with several leaf pictures in varying backgrounds.

### B. Image Processing Performance

Evaluation of the preprocessing and segmentation stages was done on the basis of clarity and accuracy.

- Image resizing and enhancement was used to enhance input quality.
- Any noise removal method minimized the unnecessary distortions.
- Segmentation- The infected regions were accurately divided into healthy areas.
- There was clear differentiation of diseased and non diseased areas.

### C. Disease Detection Results

The healthy and infected leaf images were used to test the system.

- Correct identification of healthy leaves was done.
- Detection of diseased leaves was done using changes in color and texture.
- Symptoms of common diseases (spots and discoloration) were also identified.
- The total accuracy in the detection of test images was satisfactory.

### D. Infection Percentage Calculation

The severity is computed with the help of the pixel analysis within the system.

- Infected pixels were counted correctly.
- Total leaf pixels were calculated properly.
- Percentage of infection calculated based on formula.
- Low, Moderate, and High severity levels.

### E. Remedy Recommendation Results

According to identified illness:

- The use of organic solutions like neem oil and leaf extract were proposed.
- Pesticides and fungicides inorganic remedies were supplied.
- Suggestions were pertinent to the identified disease.

### F. System Performance

- Image processing time: 2-5s.
- Works with various leaf images and backgrounds.
- Gives proper and comprehensible output.
- Light and easy to navigate.

### G. Output Results

The system displays:

- Detected disease name
- Infection percentage
- Severity level
- Organic and inorganic remedies

### H. Application Screenshots

**Figure 1 : Leaf Image**

The input leaf image, which is fed into the system is demonstrated in the above figure. OpenCV is used to load the image in the given location. This image is the primary input of the disease detection process.


**Figure 2: Disease Percentage**

This number is used to show the first analysis of the image of the leaf under which the system starts to distinguish the areas of infection. The system examines the color and texture changes that occur on the surface of the leaf in order to identify potential areas of disease.

**Figure 3: Infected Area of Leaf**

This image depicts the infected area of the leaf as detected following the image segmentation. This is achieved by a system that separates the diseased area of the leaf as compared to the healthy part of it.

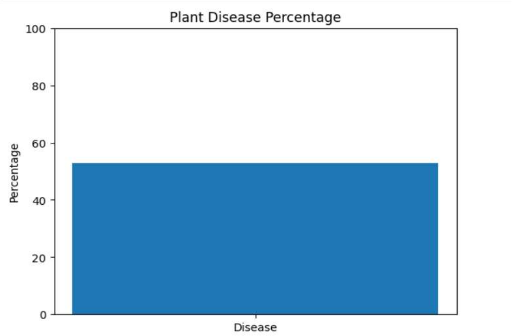



**Figure 4 : Disease Percentage**

This number shows the percentage of infection obtained via pixel analysis. The system compares the number of pixels that are infected to the total number of pixels of the leaves to estimate the severity of the disease.

Disease Detected: Leaf Blight

Recommended Remedies:

✓ Remove infected leaves immediately.
✓ Apply copper-based fungicides.
✓ Avoid overhead watering.
✓ Improve air circulation around plants.

**Figure 6 : Disease Detection**

This figure shows the disease that was identified in the leaf following analysis. Image processing techniques are used to determine whether the leaf is healthy or the disease that it is infected with.

Disease Detected: Leaf Spot (Fungal)

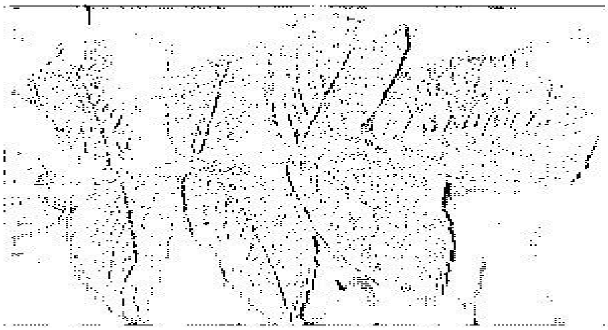
Organic Remedies:

✓ Spray Neem oil (5 ml per 1 liter of water) weekly.
✓ Do not water leaves directly; water only at soil level.
✓ Improve sunlight and airflow around plant.
✓ Remove infected leaves to stop spreading.

Additional Advice: Mulching and compost tea can boost plant immunity.

**Figure 7 : Organic Remedies**

This figure indicates the proposed organic solution to the control of the identified plant disease. These home remedies would involve natural remedies like neem oil spray or pulling off the infected leaves.



Disease: Leaf Spot

HOW the disease occurs:

- Fungal spores land on leaf surface
- Spores germinate under moist conditions
- Infection spreads to healthy tissues

WHY the disease occurs:

- High humidity around the plant
- Water droplets remain on leaves
- Poor air circulation
- Warm and wet climate

WHEN (time) it occurs:

- Occurs mostly during rainy seasons
- More common in early monsoon
- Can increase after frequent watering

Disease information not available.

**Figure 8 : Reasons for Disease**

This number describes the potential causes of the development of the disease. It also gives data on time and mode of occurrence of the disease in plants.

1 Disease Percentage: 52.99%
2 Disease Level: High Disease

**Figure 9 : Results**

*this figure displays the final output of the system. it shows the detected disease name, infection percentage, and recommended organic and inorganic remedieshh for controlling the disease.*

## V. Conclusion

The proposed project involves a highly effective and dependable system of plant leaf diseases detection through image processing and machine learning tools. The system is effective in analyzing leaf images so as to identify diseases using the color, texture, and shape characteristics. It also determines the percentage of infected area and is useful in the determination of the degree of severity of the disease and gives a better insight into the state of the plant.

The accuracy of disease detection is enhanced with the implementation of image preprocessing, segmentation, and feature extraction. The system can distinguish the regions of the infection and determine the percentage of the infection with the help of a pixel analysis. Besides, the system proposes appropriate remedies, both organic and inorganic which is useful in real world agriculture.

The proposed system will minimize human error, manual efforts, and also enable farmers to take timely preventive measures to manage the spread of diseases. It promotes early diagnosis, enhances crop yield, and is a factor to sustainable farming. In general, this system is a very easy to use, economical, and effective system in monitoring and controlling plant diseases.

## VI. Future Work

The proposed system can also be improved in the future by creating a mobile application that will enable users to take the pictures of leaves on a smartphone camera in real time. Models of advanced deep learning like Convolutional Neural Networks (CNN) can be incorporated to enhance precision and efficiency of the disease detection in various environmental conditions. The system can further be expanded to accommodate more plant species and diseases. Also, the combination of the cloud-based storage and IoT should allow monitoring the health of crops continuously. To make the system more realistic and helpful in the modern smart agriculture, it is possible to add real-time notifications and more specific remedy recommendations.

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